

Have you tried asking Codex/Claude to call the other in headless mode to review its work?

You should install both codex and agy cli

If you have the shared drive, as long as it is synced to your device, you don't even need the MCP

But you still want to de-conflict

so like one shared google drive link and it will have one data folder and then one folder for me, one folder for you etc

And give it screenshots in IDE or browser as it walks you thru the process or whenever you get stuck and it will get you up and running in no time.

I am sure IDE is fine too but I don't know anything about it and I hear there are functions that don't transport over.

You want to be in CLI anyway, so good to switch today

Okay

Let me lay out the high-level objectives this week for everyone.

I am super-stressed for time and will only be giving high-level instructions along the way. It is your responsibility to plan, flag key moments of your plans to me, trouble-shoot together, and raise your hand and get my attention when you need it

Things often move very fast and very fluidly these days, but also I will be pulled into tv shows or other things and won't be very hands on after today.

1) I want a systematic ingestion/processing of past economic surveys to put together a definitive time series for every statistic we can find this week.

There is a lot of rich data in the econ surveys and the budget, and I want us to have a gold-standard database of everything.

ECONOMIC AND

INDICATORS	1960s	1970s	1980s	1990s	2000s	2007-08	2008-09	2009-10	2010-11
	Average (Annual)								
FINANCIAL SECTOR:									
GROWTH RATE (at constant fc) %									
GDP	6.8	4.8	6.5	4.6	4.5	5.0	0.4	2.6	3.6
Agriculture	5.1	2.4	5.4	4.4	2.8	1.8	3.5	0.2	2.0
Manufacturing	9.9	5.5	8.2	4.8	7.3	6.1	-4.2	1.4	2.5
Commodity Producing Sector	6.8	3.9	6.5	4.6	4.8	5.1	-0.9	1.8	3.2
Services Sector	6.7	6.3	6.7	4.6	5.2	4.9	1.3	3.2	3.9
GROWTH RATES (at current mp) %									
Total Investment	-	21.8	4.2	8.1	15.6	17.7	13.4	1.4	9.8
Fixed Investment	14.8	20.5	3.7	7.8	15.7	17.9	12.4	0.3	8.4
Public Investment	14.0	25.3	2.6	7.3	12.5	21.0	11.2	-2.1	6.6
(including general govt.)									
Private Investment	20.9	17.0	5.1	8.8	17.5	16.8	12.9	1.2	9.0
(as % of Total Investment)			5.1						
National Savings	-	67.5	79.2	75.4	89.9	57.5	68.6	85.9	100.7
Foreign Savings	-	32.5	20.8	24.6	10.1	42.5	31.4	14.1	-0.7
(as % of GDP current mp)									
Total Investment	-	17.1	18.7	18.3	17.9	19.2	17.5	15.8	14.1
Fixed Investment	-	15.9	17.0	16.6	16.4	17.6	15.9	14.2	12.5
Public Investment	-	10.3	9.2	7.5	4.6	4.8	4.3	3.7	3.2
Private Investment	-	5.6	7.8	9.1	11.8	12.8	11.7	10.5	9.3
National Savings	-	11.2	14.8	13.8	15.9	11.0	12.0	13.6	14.2
Foreign Savings*	-	5.8	3.9	4.5	2.0	8.2	5.5	2.2	-0.1
Domestic Savings	-	7.4	7.7	14.0	14.6	9.1	9.4	9.8	9.7
Per Capita Income (mp-US \$)*	-	-	-	-	746.0	1053.2	1026.1	1072.4	1274.1
GDP DEFLATOR (growth %)	-	-	2.3	8.3	8.4	12.9	20.7	10.7	19.5
CONSUMER PRICE INDEX (CPI)									
(growth %)	3.2	12.5	7.2	9.7	7.3	12.0	17.0	10.1	13.7
FISCAL POLICY									
(as % of GDP mp)									
Total Revenue	13.1	16.8	17.3	17.1	13.9	14.1	14.0	14.0	12.3
Tax Revenue	-	-	13.8	13.4	10.3	9.9	9.1	9.9	9.3
Non-Tax Revenue	-	-	3.5	3.7	3.6	4.2	4.9	4.1	3.0
Total Expenditure	11.6	21.5	24.9	24.1	18.3	21.4	19.2	20.2	18.9
Current Expenditure	-	-	17.6	19.4	15.1	17.4	15.5	16.0	15.9
Defence	-	-	6.5	5.6	3.1	2.6	2.5	2.5	2.5
Markup Payments	-	-	3.8	6.8	4.9	4.8	5.0	4.4	3.9
Others	-	-	7.3	7.0	7.2	10.0	8.0	9.2	9.6
Development Expenditure	-	-	7.3	4.7	3.3	4.0	3.5	4.4	2.8
Overall Deficit	2.1	5.3	7.1	6.9	4.4	7.3	5.2	6.2	6.5
MONEY & CREDIT (growth %)									
Monetary Assets (M2)	16.3	21.0	13.2	16.8	15.0	15.3	9.6	12.5	15.9
Domestic Assets	15.0	20.5	15.4	12.2	14.1	33.6	15.4	12.7	13.1
STOCK EXCHANGE (growth %)									
KSE 100 Index	-	-	0.1	4.1	27.2	-10.8	-41.7	35.7	28.5
Aggregate Market Capitalization	-	-	2.5	13.4	29.1	-6.0	-43.9	28.8	20.3

- : Not available mp : Market prices fc : Factor cost P: Provisional, R: Revised, F: Final

*: At average exchange rate used in National Accounts Committee meeting

**: July-April

Note: From 2016-17, CPI is estimated on 2015-16=100 as base year

Stuff like this has to graduate from collecting dust in PDFs to being always available

🚩 It is absolutely essential in this process for our work to be completely reproducible 🚩

https://drive.google.com/drive/folders/1a-R9JW43nKeoKxIZFoW0wHfWTDuIKaaf?usp=share_link

That means that all data downloads have to be recorded (i.e. URL etc) and saved in original form, and all manipulation must be done with recorded code on copies of the data

I have enabled edit access for everyone

Make original data in the repository read-only for all except me and you

Reproducibility means that anything you do have to be done in a way that you can give the "recipe" to someone else and they can reproduce and verify

When you tell your claudes and codexes to ensure reproducibility, they will be good about ensuring it.

So the first thing is to just build out our time-series as much as possible by ingesting the documents, scraping data, and adding to a database, and then simply keep doing over and over for each year we have until our repository is as complete as possible.

2nd, we want to absorb and keep track of what the commentary said at what time.

a) Important Crops

i) Cotton

Being a major cash crop of Pakistan, cotton is considered the backbone of the economy. It contributes about 0.8 percent to GDP and 4.1 percent of total value addition in agriculture. During 2019-20, cotton crop was sown on an area of 2,527 thousand hectares, which increased by 6.5 percent over last year's area (2,373 thousand hectares). Cotton production is estimated at around 9.178 million bales, which is lower by 6.9 percent over the last year's production of 9.861 million bales (see Table 2.4 and Figure 2.1). Even though the overall area increased over last year, the overall performance remained below due to unfavourable weather and low water availability during important stages of plant development along with pest attacks.

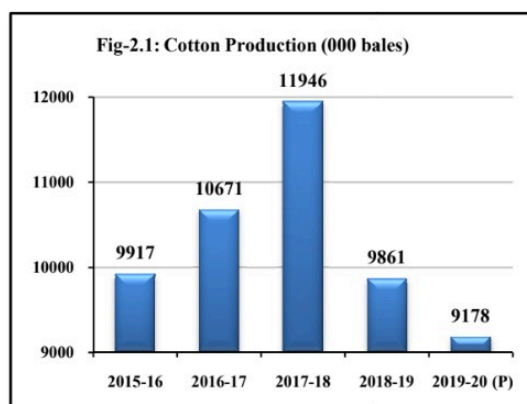


Table 2.4: Area, Production and Yield of Cotton

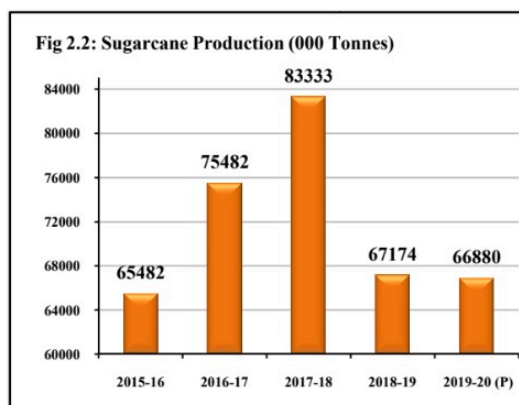
Year	Area		Production		Yield	
	(000 Hectare)	% Change	(000 Bales)	% Change	(Kgs/Hec)	% Change
2015-16	2,902	-	9,917	-	582	-
2016-17	2,489	-14.2	10,671	7.6	730	25.3
2017-18	2,700	8.5	11,946	11.9	753	3.1
2018-19	2,373	-12.1	9,861	-17.5	707	-6.1
2019-20(P)	2,527	6.5	9,178	-6.9	618	-12.6

P: Provisional (July-March)

Source: Pakistan Bureau of Statistics

ii) Sugarcane

Sugarcane is a high value cash crop that has significance for sugar and sugar related industries in Pakistan. It contributes about 0.6 percent to GDP and 2.9 percent of total value addition in agriculture. During 2019-20, sugarcane production decreased by 0.4 percent to 66.880 million tonnes as compared to 67.174 million tonnes of last year. The output may depict the pattern of area under cultivation i.e 1,040 thousand hectares compared to 1,102 thousand hectares, a decline of 5.6 percent, with improved yields compared to 2018-19. The area, production and yield of sugarcane during the last five years are given in Table 2.5 and Figure 2.2.



Here's a random page to illustrate this.

You have to ask your tools to enter plan mode and think about how to create usable summaries from these

The problem with AI is that it has limited short-term attention (i.e. context), and PDFs are much heavier for it to use than .md files

So we need it to ingest different documents and develop summaries.

The problem with summaries is that by definition, they involve signal-loss: we will lose subtleties etc.

That is fine. We can't know everything at all times.

But something for us to know.

Data will be in our databases and be readily and fully available.

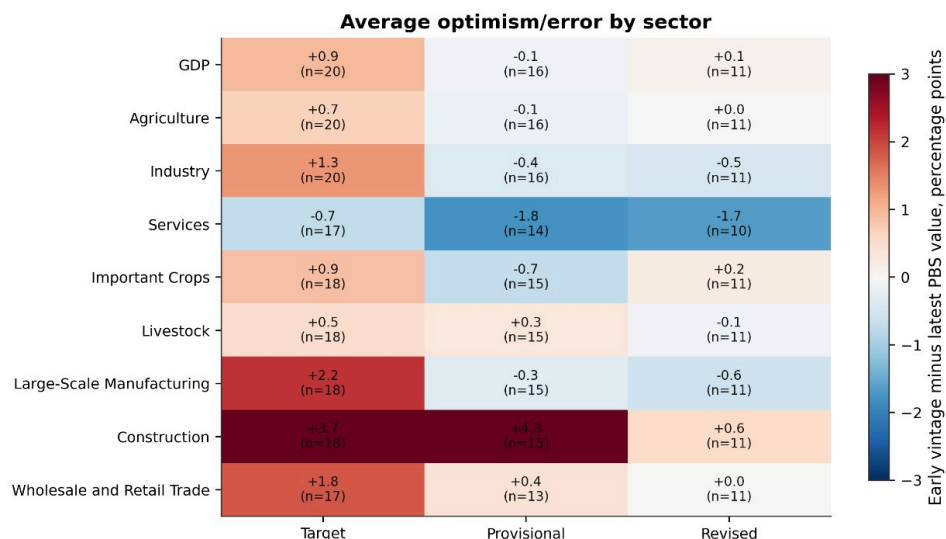
Commentary will only be usable in summaries.

One option is to ask it to pick out different narratives and create multiple summaries.

For example, here, we can ask it to just make a list of cause-effect claims, and not any actual data in the summaries.

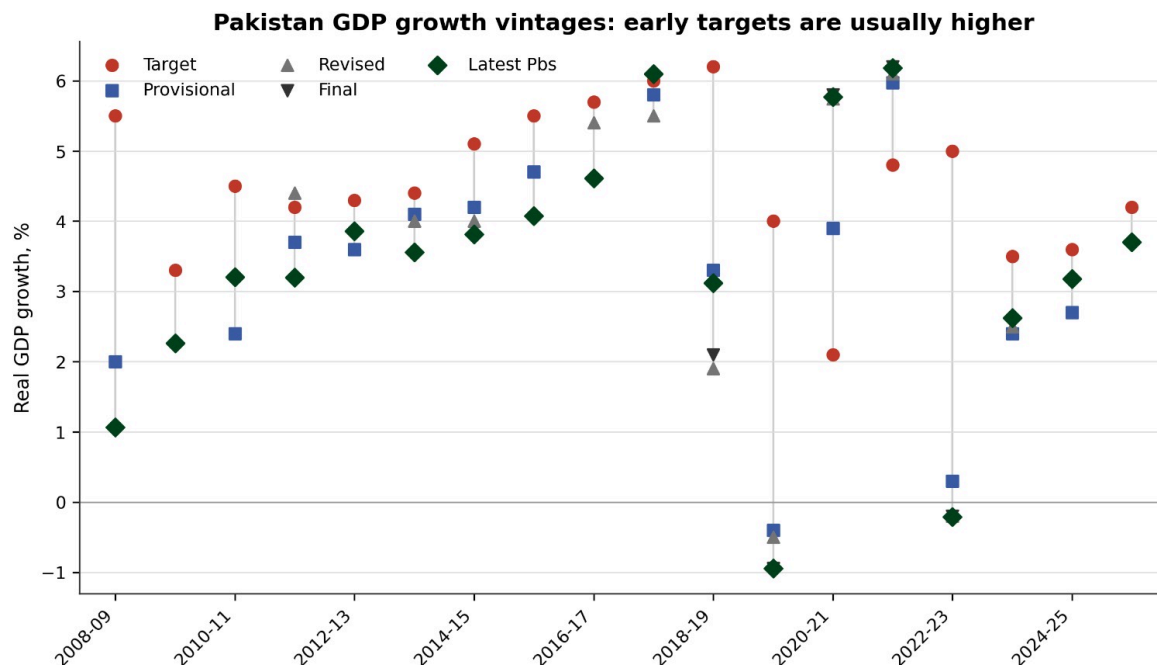
The higher level objective here is to simply keep track and be able to comment on the type of commentary we have had.

To give you an example:



Source: Planning Commission Annual Plans; PBS National Accounts Annual Tables 2025-26, Table 6. Positive cells mean the early vintage was higher than the latest PBS vintage.

Yesterday I asked Codex to track provisional versus final revised economics data across multiple years and it built this.



Source: Planning Commission Annual Plans; PBS National Accounts Annual Tables 2025-26, Table 6. Latest PBS = latest restated series at constant basic prices of 2015-16.

As you can see, our economics commentary is very bad in 2008 (GFC), 2018 (Dar), 2020 (COVID), and 2022 (VONC). These are the big endogenous or exogenous shock years.

And for other years, there is a small but consistent optimism. So I can be reasonably confident that I need to trim actual expectations compared to the data being shared by the average commentator just on face value.

Now: a few months ago I had AI build a graph, and it came out looking all beautiful and impressive like this, and it was complete hallucination.

If we publish even a single figure that is substantially off, my/LUMS' reputation gets shot. That is death. Completely unacceptable.

So how do we guard against it?

Multiple things

First, our reproducibility rules come in to help. We track and trace everything we did.

We never ask AI to build graphs from scratch

We download gold-standard data from official sources.

We make AI build figures.

We hand off its work to a fresh-context different model for adversarial review and reproduction (i.e. Claude Code output gets handed to Codex, Codex output to Agy, Agy to Claude etc)

Two weeks ago, I did all of this and tweeted a graph that Taimur Jhagra retweeted, and then I nearly died of embarrassment

Because I had done everything above

But the gov had reported GDP figures, and then said something about income per capita

So I figured that meant GDP per capita and I found a "mistake"

And both Claude and Codex independently said you have found a mistake

So I published it

And then a friend called and said the data was GNI per capita and I was mistaken

So I had to retract

This is unacceptable, and so the lesson learned was that I have to pay attention to definitions and data series construction, and should pass output to AI to search for how well at least parts of the graph match with past graphs reported in other sources.

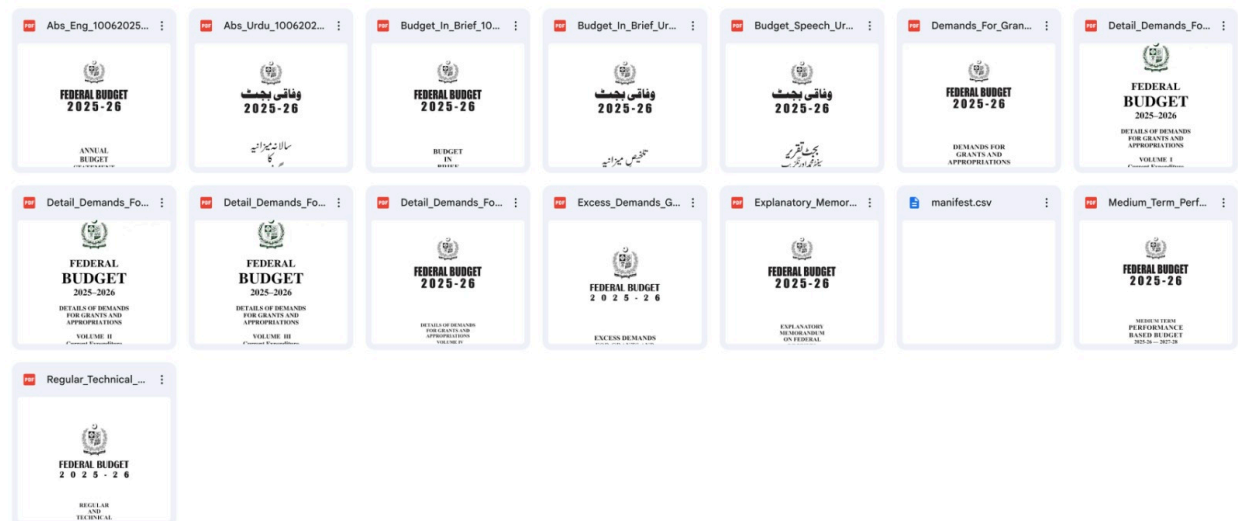
So just to recap so far before I go on:

1 - Ingest and have *every* time series in Pakistan

2 - Do ingestion and summaries from all economic surveys, remembering that summaries can entail signal loss.

* Reproducibility is king, and AI workflows need strong guardrails *

🔥 3 - I want to get deeper into budget documents than ever before 🔥



The budget is all of these documents, and they will get released one by one To explain further, let me send a voice note on this

I am still trying to understand all the ways in which these documents connect to each other for a given year too.

@9818412728528 why don't you take this on. Make a copy of one year of all budget docs in a new folder, and then open Claude Code in it and make it review and surface connections and tell us how to approach this.

#3 is us having the most intellectually honest and professional reactions to this year's budget of anyone in Pakistan, and producing outputs in near-real time.

Most likely, the budget is not coming on Friday, and is now delayed to some day in the 8th to 12th bracket.

This gives us time to really think thru what will get released when, and what analysis we want to do on it.

One way to do this is to analyse everything up to say 2023 first. And then ingest one year of docs one by one and flag what new analysis emerges. Maybe review what analysis got done for that year. And then move on to repeating and refining the approach for 2024, and finally for 2025.

So I want that as soon as the new docs drop in our folders, or even if we get screenshots or pdfs or jpgs of scraps of info, our system is primed to quickly find and develop the best form of the analysis/report and publish.

Now @24004723269638, I am assuming you already have #1 mostly sorted. Please give me a sitrep on that specifically.

@197392586563804 why don't you start #2 and keep me closely informed as you go And @24004723269638 I need to know about the state of the bigger project and what we need to do to get it launched ASAP

Yes sir number 1 is mostly sorted, the db and the backend are setup which means that importing the economic survey data is not going to be a new task, we already have a system for importing them in

for the time being, i am acting as db admin which means the responsibility of importing data into the site becomes my core responsibility

then comes the site itself which consists of two pieces, the front end or backend

again, lets not go too deep into that, i will be briefing Ibrahim deeply on this so that he understands later on, we have been iterating on frontends

@9818412728528 when you are done with the budget, which is your main priority, you can start thinking about #4, which is to record and review the biases and contradictions in the "plans" put forward by different interest groups.

Your summary is an excellent first overview. You may also want to ingest recent IMF reports and see how they might differ.

In the next iteration, I will want you to think through different cuts at the docs - asking Claude to generate different types of summaries extracting different stuff from each report - and to develop overview maps.

But this is strictly after #3

Let me continue this in DM with you, and let's reserve this channel exclusively for budget analysis for the next 10 days.

perfect sir, will look into this after the first task.